LE TRUNG KIEN NGUYEN (HE/HIM)

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EDUCATION

University of South Florida — **Tampa, FL**
B.S. Mechanical Engineering, Expected May 2027 | GPA: 3.84 — *08/2024 – present*

TECHNICAL SKILLS

- **CAD / Design:** SolidWorks (CSWA), part design, surface modeling, mechanical assemblies, engineering drawings, actuator components, DFM for 3D printing, McMaster-Carr sourcing, GD&T basics
- **Analysis / Simulation:** Ansys, SolidWorks Simulation, topology optimization, first-principles stress/strain analysis, torque sizing, tolerance analysis, stiffness estimation
- **Programming / Test:** MATLAB, Python, C/C++, Arduino IDE, Visual Studio Code, data analysis, test scripting
- **Robotics / Actuators:** Planetary gearbox prototypes, cycloidal gearbox for NEMA 23 servo targeting 10 Nm torque, motors, bearings, sensors, inverse kinematics, basic motion control
- **Manufacturing:** 3D printing, Bambu Lab printers, soldering, hand tools, shop tools, prototype assembly, fit-checking

EXPERIENCE

VinMotion — **Hanoi, Vietnam**
Mechanical Engineering Intern – Mechanical Design, CAD, Prototyping & Assembly — *06/2025 – 08/2025*

- Inspected 19 humanoid robot assemblies across 16 A1.6 models and 3 A2.1 models by performing quality checks, assembly reviews, and failure reports, helping the team identify design errors before public product exhibition.
- Built 2 full-scale 3D-printed duo-planetary actuator prototypes with the mechanical and actuator teams by supporting CAD review, part fit-checking, and prototype assembly, enabling future experiments for self-manufactured actuator designs.
- Improved actuator and robot assembly reliability by reporting design failures, fixing CAD-related issues, and recommending mechanical design changes to senior engineers.
- Standardized assembly knowledge for future interns by creating A1.6 and A2.1 assembly instructions, reducing onboarding time and helping new team members understand assembly procedures faster.

NASA Student Launch 2025 — **Tampa, USA**
Aerostructures & Airbrake Mechanical Design Junior Engineer — *09/2024 – present*

- Improved rocket airbrake drag performance by approximately 200% by designing and iterating 3 SolidWorks prototypes for the Aerostructure and Airbrake subteam.
- Designed a compact airbrake mechanism targeting 42 N drag per flap and ≤4 lb system mass by optimizing lever-arm travel, packaging, and prototype geometry for subsonic flight conditions.
- Reduced damaged-fin replacement time by approximately 50% by designing a bolted-fin frame in SolidWorks for the full-scale rocket.
- Supported PID calibration and flight-readiness testing by measuring assembly dimensions, mass properties, and component angles during rocket integration.
- Manufactured and assembled custom rocket components using hand tools and shop equipment, resolving fit and assembly issues during prototype development.

ViSEF and ISEF — **Texas, USA**
Researcher — *09/2022 – 05/2023*

- Reduced quadruped robot foot pressure by approximately 25x by designing a bio-inspired webbed foot with increased end-effector contact area for muddy landslide terrain.
- Improved robot stability on mud with approximately 50% water composition by validating webbed feet against rounded feet, reducing measured submersion from 35/21/20/10 mm to 10/6/5/3 mm across the four legs.
- Enabled terrain-aware foot placement by integrating FSR force sensors and analyzing force feedback from hard ground and muddy ground conditions.
- Demonstrated multi-gait mobility by programming and testing 8 robot gaits, including forward walking, turning, pitch, roll, and yaw motions.
- Validated slope-climbing capability by testing the quadruped robot on a 20° hard-surface incline while carrying a 1 kg rescue payload.
- Selected servo motors using first-principles torque calculations based on leg geometry, stance, robot mass, and joint-angle conditions.

LEADERSHIP & AWARDS

- Head of Technical Faculty, Le Hong Phong Science Club; led technical teams for robotics competitions across automation and vehicle-design categories.
- Awards: USAID Special Award at ISEF 2023; 1st Prize ViSEF 2023; 1st Prize Technion Robotraffic 2023.